

Hand Eye calibration for industrial robotic systems

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1 Introduction

The integration of machine vision and industrial robotics has become increasingly important in modern manufacturing systems. Applications such as automated inspection, object localization, quality control, and vision-guided manipulation require an accurate geometric relationship between the robot and the camera. Any error in this relationship directly affects the accuracy and reliability of the overall system.

To establish this relationship, a calibration process is required. In particular, hand-eye calibration determines the rigid transformation between the camera reference frame and the robot end-effector or robot base frame. Although numerous calibration algorithms have been proposed in the literature, their practical performance is strongly influenced by factors such as image quality, calibration target geometry, motion diversity, and measurement noise. The objective of this thesis was the design, implementation, and validation of a complete vision-based calibration framework for industrial robotic systems. The developed solution integrates camera intrinsic calibration, pose estimation using ChArUco calibration boards, and hand-eye calibration within a unified software pipeline developed in Python using OpenCV. Particular attention was devoted to evaluating the influence of dataset quality, acquisition strategy, and algorithm selection on the final calibration accuracy.

In addition to the experimental validation performed in a laboratory environment, the knowledge acquired during the project was applied to the development of a calibration module integrated into an industrial vision-guided robotic system deployed on a production machine manufactured by EPF.

2 Calibration Methodology

The proposed calibration framework consists of three main stages: intrinsic camera calibration, target pose estimation, and hand-eye calibration.

The intrinsic calibration stage estimates the camera matrix and lens distortion coefficients using multiple observations of a ChArUco calibration board. ChArUco boards combine the robustness of ArUco markers with the geometric precision of chessboard corners, providing reliable feature detection even under partial occlusions. The estimated intrinsic parameters are subsequently used to correct lens distortion and enable accurate pose estimation.

Once the camera parameters are known, the pose of the calibration target relative to the camera is computed using a Perspective-n-Point (PnP) approach. For each acquired image, the detected ChArUco corners are matched with their corresponding three-dimensional coordinates on the calibration board, allowing the estimation of the camera-to-target transformation.

The final stage consists of hand-eye calibration. Several algorithms available in OpenCV were investigated, including the methods proposed by Tsai and Lenz, Park and Martin, Horaud, Andreff, and Daniilidis. These methods solve the classical hand-eye calibration problem expressed by the equation

$$AX = XB$$

where A represents the robot motion between two poses, B represents the corresponding camera motion, and X is the unknown rigid transformation between the robot and camera reference frames. To improve calibration robustness, additional refinement techniques were introduced. These include reprojection-error-based filtering of invalid observations, systematic pose selection to maximize motion diversity, and geometric consistency analysis using the Kabsch–Umeyama rigid alignment method. Experimental investigations showed that calibration quality depends more strongly on the quality and diversity of the acquired data than on the specific hand-eye algorithm itself.

3 Experimental Evaluation

The experimental validation was performed using a FANUC LR Mate 200iD industrial robot equipped with a Cognex industrial camera operating in an eye-in-hand configuration. A ChArUco calibration board was rigidly positioned within the robot workspace and observed from multiple viewpoints. Two industrial cameras were evaluated during the intrinsic calibration phase. Preliminary experiments performed using a 1280×1024 pixel camera achieved a reprojection error of approximately 1.24 pixels. To improve localization accuracy, the camera was replaced with a 12 MP Cognex sensor providing images at 4096×3000 pixels. The higher image resolution significantly improved corner localization accuracy and reduced the reprojection error to approximately 0.68 pixels.

A noteworthy aspect of the experimental setup was the use of a lens exhibiting extremely strong optical distortion, comparable to that encountered in wide-angle imaging systems. Distortions near the image boundaries reached several centimetres before correction. Despite these challenging conditions, the intrinsic calibration procedure successfully estimated stable camera parameters and produced geometrically consistent undistorted images, demonstrating the robustness of the adopted approach.

The influence of dataset size on hand-eye calibration accuracy was investigated through a convergence analysis. Starting from a small number of observations, additional image-pose pairs were progressively included in the calibration process. The resulting transformations were evaluated using a cross-image consistency metric based on the reconstructed three-dimensional positions of ChArUco corners expressed in the robot base frame.

The results showed that calibration accuracy improved significantly as additional observations were incorporated. The consistency error decreased rapidly during the initial acquisition phase and converged after approximately fifteen observations, indicating

that larger datasets provide diminishing returns beyond a certain point. This behaviour confirms the importance of acquiring sufficiently diverse viewpoints while also providing practical guidelines for future calibration procedures.

A comparison of the different hand-eye calibration algorithms revealed that all evaluated methods achieved sub-millimetre consistency errors. The Tsai, Park, and Horaud methods demonstrated the most stable behaviour across different datasets, while Andreff and Daniilidis exhibited greater sensitivity to dataset characteristics. Overall, the results indicate that multiple calibration methods can achieve highly accurate solutions when supported by high-quality data and appropriate acquisition strategies.

The experimental analysis also highlighted several practical sources of error. These include the limited depth observability associated with planar calibration targets, motion degeneracy resulting from insufficient pose diversity, image acquisition noise, and systematic geometric biases introduced by imperfect target positioning. In particular, a slight inclination of the calibration board was identified through reconstruction analysis and shown to affect the consistency of the estimated transformations.

4 Industrial Deployment and Conclusions

Beyond the laboratory validation, the methodologies developed during this thesis were applied to the implementation of a complete eye-to-hand calibration module for a SUPATA automatic feeding machine manufactured by EPF. The system employs a fixed overhead camera and a KUKA industrial robot to identify, localize, and pick plastic cable connectors from a vibrating tray. The developed calibration module replaced a previous solution that exhibited limited robustness, particularly near the boundaries of the camera field of view. The new implementation was integrated into the existing production software architecture and deployed on a customer-delivered machine.

Operational validation was performed under normal production conditions. The system successfully executed several tens of thousands of pick-and-place cycles over approximately three days of operation without calibration-related failures. Although a formal metrological accuracy assessment was not available in the production environment, the successful deployment provided practical evidence of the reliability and industrial applicability of the developed calibration methodology.

In conclusion, this thesis demonstrates that accurate and reliable vision-based calibration can be achieved using low-cost calibration targets and widely available computer vision tools, provided that sufficient attention is devoted to data acquisition, motion diversity, and geometric consistency. The proposed framework was validated both experimentally and in an industrial production environment, confirming its suitability for real-world robotic applications.